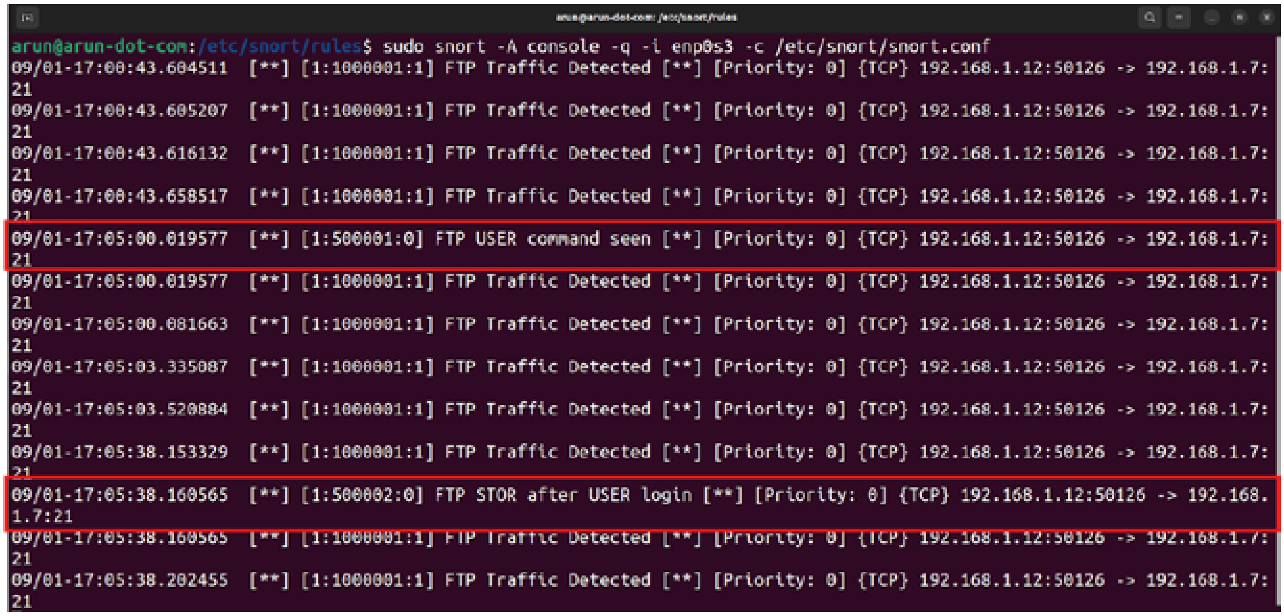

Figure 8: Successful listing of alert messages in the Snort console

the Ubuntu system using the IP address of another machine as follows:

```
$ ftp192.168.1.7
```

When prompted, the user should enter a valid username and password to gain access to the FTP server. After logging in, the user can upload a file using the *put* command. Even if the user does not have permission to access the file contents, the command will succeed, and the FTP activity can be observed in the Snort console.

The expected outcomes are:

- The *ftp* command should work successfully, and the user should be prompted for user details.
- The Snort console should display alerts for FTP traffic.

Once the user details are prompted, the FTP traffic configured with dynamic rules in Snort will start working.

The expected outcomes are:

- When the user enters the *USER* command, Snort should detect the login attempt and trigger the first dynamic rule.
- When the *put* command is used to upload a file, Snort should detect the *STOR* command and trigger the second dynamic rule.
- Both alerts should appear in the Snort console, showing that the multi-step dynamic detection is working as intended.

The alerts generated by the Snort console are also stored in the Snort log file. This log file is stored in */var/log/snort/snort.alert.fast*. Each alert entry includes details such as the timestamp, source and destination IP addresses, the rule that was triggered, and a brief description of the detected activity. Reviewing this log allows verification that Snort correctly detected and logged the actions.

By the successful listing of commands in the Snort installed terminal, it can be confirmed that the IDS installation and setup have been completed successfully.

Snort plays an important role in securing networks by acting as a powerful intrusion detection system. It constantly watches the flow of traffic and quickly identifies unusual or suspicious activity that could harm the system. With its open source nature and signature-based detection, Snort has become a widely trusted tool for both learning and real-world use.

This is how you can set up an intrusion detection system (IDS) on Ubuntu using Snort. The steps include preparing the system, installing Snort, changing the main settings, and adding custom rules to detect FTP traffic. The setup can then be tested by sending traffic from another system, and Snort can monitor it and show alerts. With this IDS running, the network becomes safer, suspicious activity can be noticed quickly, and the system is ready for more security improvements in the future. END

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